TechToons — Product Requirements Document

Product: TechToons
Date: February 23, 2026
Author: Product Team
Version: 1.1
Status: Updated — v0.2
Launch Market: KE Kenya (East Africa)

1. Executive Summary

TechToons is an interactive, browser-based coloring book that introduces children (ages 4–10) to technology concepts through creative play. Children color tech-themed illustrations — robots, circuit boards, AI brains, rockets, drones, and VR headsets — using intuitive digital tools, building familiarity with STEM imagery while exercising creativity.

Launch Market: Kenya is chosen as the initial market due to its rapidly growing EdTech sector, high mobile internet penetration (~90% of internet access via mobile), a young population (over 40% under age 15), and the Kenyan government's emphasis on digital literacy through the CBC (Competency-Based Curriculum) framework.

Business Goal: Transform the existing prototype into a commercially viable EdTech product targeting Kenyan families and schools, generating recurring revenue through an affordable freemium model priced for the Kenyan market, before expanding across East Africa and beyond.

2. Problem Statement

The Gap

Kenyan parents and educators seek engaging, screen-based activities that are both creative and educational. With the rollout of Kenya's CBC curriculum emphasizing digital literacy and creative skills, the demand for locally relevant EdTech tools is surging:

PAIN POINT	KENYA CONTEXT	TECHTOONS OPPORTUNITY
Screen guilt	Parents worry about passive screen time on phones	Active creation builds productive screen habits
STEM exposure gap	CBC mandates digital literacy, but few engaging tools exist	Introduces tech concepts at age 4–10 through play
Costly alternatives	Global apps (Toca Boca, ABCmouse) are priced for Western markets	Affordable pricing in KES via M-Pesa
Generic content	Most coloring apps use Western-centric themes	Tech-themed illustrations with future local content
No progress tracking	Parents and teachers can't see engagement data	Dashboard shows time, pages, and skills
Connectivity gaps	Many areas have intermittent internet	Lightweight SVGs work on low-bandwidth connections

Target Users

PERSONA	DESCRIPTION	NEEDS
Kids (4–10)	Kenyan primary school children (Grade 1–5)	Fun, intuitive, rewarding experience on any device
Parents	Urban & peri-urban Kenyan parents with smartphones	Affordable, educational, ad-free, M-Pesa payment
Educators	CBC teachers in Kenyan primary schools	CBC-aligned activities, class management, works on school devices

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3. Product Vision & Success Metrics

> *"Make every child's first interaction with technology a creative one."*

Key Results (12-month targets)

METRIC	TARGET	MEASUREMENT
Monthly Active Users (MAU)	30,000 (Kenya)	Analytics
Free → Paid Conversion	4%	Subscription data
Monthly Recurring Revenue (MRR)	KES 600,000 (~$4,600)	M-Pesa / Stripe dashboard
Average Session Duration	8+ minutes	Analytics
Google Play Rating	4.5+ stars	Play Store reviews
Kenyan Schools Onboarded	200	Admin dashboard
Pages Colored (total)	500,000	Event tracking

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4. Current Product Capabilities (v0.2 — Enhanced Prototype)

The current build demonstrates core viability plus kid-friendly UX:

Core Features

- ✅ 6 tech-themed SVG coloring pages (Robot, Circuit Board, AI Brain, Rocket, Drone, VR Headset)
- ✅ Click-to-fill region painting with inline SVGs
- ✅ 16-color curated palette + full-spectrum custom picker
- ✅ 4 brush tools: Paint Bucket (fill), Freehand Brush (draw with finger/stylus), Eraser, Rainbow (random)
- ✅ 20-level undo stack + full reset
- ✅ Export to PNG (2× resolution)
- ✅ Responsive layout (desktop + mobile)
- ✅ Clean design system (Fredoka + Nunito, teal/coral palette, 8px grid)
- ✅ Deployed on Vercel

Kid-Friendly UX NEW IN V0.2

- ✅ **Labeled Toolbar** — Visible text labels on every toolbar control ("Paint", "Erase", "Rainbow", "Undo", "Start Over", "Save") with section headings ("Colors", "Tools", "Actions")
- ✅ **Live Region Label Indicator** — Dynamic indicator above the canvas ("🎨 You're coloring: Head") that updates on hover/click, so kids know exactly what part they're coloring

- ✅ **Region Label Maps** — 100+ human-readable region names across all 6 pages (e.g., "Antenna Ball", "Processor (CPU)", "Left Thinking Area", "Nose Cone", "Camera Lens", "Left Controller")
- ✅ **Educational Descriptions** — Each coloring page shows an age-appropriate description explaining the tech concept (e.g., "A robot has a head, arms, and legs — just like you! Can you color each part?")
- ✅ **Gallery Descriptions** — Home page cards show educational blurbs so kids understand what they'll learn before clicking
- ✅ **Guided Onboarding Prompt** — Default state shows "👆 Tap a part to start coloring!" to guide new users
- ✅ **Color Name Labels** — Selected color displays its name ("Red", "Teal", "Violet", etc.) next to the color preview for learning

Tech Stack

- React 19 + TypeScript
- Vite 7 (dev server & bundler)
- react-colorful (color picker)
- react-router-dom v7 (client-side routing)

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5. Feature Requirements

5.1 Phase 1 — Foundation (Months 1–3)

Goal: Transform prototype into a launchable product

FR-1: User Accounts & Authentication P0 — Must Have

Parents create accounts; children have sub-profiles under a parent account.

- Email/password + Google OAuth sign-up
- Phone number sign-up (common in Kenya)
- Up to 4 child profiles per parent account
- Each child profile has name, avatar, age
- Child profiles have independent saved progress
- Kenya Data Protection Act (DPA 2019) compliant data collection

Acceptance Criteria: Parent can create account, add child profiles, and switch between them

FR-2: Cloud Save & Progress Persistence P0 — Must Have

Coloring progress is automatically saved and synced across devices.

- Auto-save every 30 seconds + on page leave
- Region color state stored per user per page
- Gallery of completed works per child profile
- Offline support with sync-on-reconnect

Acceptance Criteria: User can close tab, return later, and resume coloring from where they left off

FR-3: Expanded Content Library P0 — Must Have

Grow from 6 pages to 20+ across themed collections.

- New collections: Coding Creatures, Internet World, Gaming Galaxy, Green Tech
- 5 pages per collection (20+ total at launch)
- Difficulty levels: Easy (large regions), Medium, Challenging (detailed regions)
- 3 free pages + rest behind paywall

Acceptance Criteria: Content library shows collections; free users can access 3 pages, paid access all

FR-4: Freemium Subscription Model P0 — Must Have

Monetize through tiered subscription plans priced for the Kenyan market.

- **Free Tier:** 3 coloring pages, basic tools, no save/export
- **TechToons Plus (KES 250/mo ~ $1.99):** All pages, all tools, cloud save, PNG export, no ads
- **TechToons Classroom (KES 500/mo ~ $3.99):** Plus features + class management, up to 30 students, progress reports
- **M-Pesa integration** via Daraja API (primary payment in Kenya)
- Stripe for card payments
- Google Play in-app purchase for mobile
- 7-day free trial for Plus
- Annual discount (2 months free)

Acceptance Criteria: User can subscribe via M-Pesa or card, access gated content, and manage subscription

5.2 Phase 2 — Engagement & Learning (Months 4–6)

Goal: Increase retention and add educational value

FR-5: Learning Sidebars ("Did You Know?") P1 — Should Have

Each coloring page includes age-appropriate tech facts and mini-lessons.

- Collapsible sidebar with 3–5 facts per page
- Read-aloud audio for younger children
- "Quiz Me" button with simple multiple-choice
- Facts reviewed by education advisor

Acceptance Criteria: Sidebar appears on coloring page; read-aloud plays; quiz scores display

FR-6: Achievements & Rewards System P1 — Should Have

Gamified reward system to encourage exploration and completion.

- Badges: "First Robot!", "Color Explorer" (5 pages), "Tech Master" (all pages)
- Streak tracking (consecutive days)
- Virtual sticker rewards collected in a sticker book
- Celebratory animations on milestone completion

Acceptance Criteria: Badges appear in profile; streak counter visible; stickers collectible

FR-7: Parent Dashboard P1 — Should Have

Parents can monitor child engagement and learning progress.

- Time spent per session and per page
- Pages completed and collections progress
- Quiz scores and facts learned
- Weekly email summary (opt-in)
- Screen time limits (optional)

Acceptance Criteria: Dashboard displays charts and stats; email summary sends on schedule

FR-8: Sharing & Social Gallery P2 — Nice to Have

Kids can share completed artwork in a moderated community gallery.

- "Share to Gallery" button on completed works
- Moderated (AI + human review) before publishing
- Kids can "star" others' artwork (no comments for safety)
- Share link to family via email/message
- No personally identifiable info displayed

Acceptance Criteria: Artwork appears in gallery after moderation; star counts visible

FR-9: Brush Boundary Guidance ("Paint Inside the Lines") P1 — Should Have

When using the freehand brush tool, detect if the child draws outside the coloring regions and display a friendly, encouraging nudge.

- Real-time boundary detection using canvas hit-testing against SVG region paths
- Friendly animated messages: "Oops! Try coloring inside the lines 🎯" or "Almost! Stay inside the shape 😊"
- Gentle toast notification (non-blocking), auto-dismisses after 3 seconds
- Haptic feedback on mobile devices when painting outside
- Optional "Training Mode" with highlighted region borders for younger children
- Configurable sensitivity (tolerant for ages 4–5, precise for ages 8–10)
- No penalty — purely guidance, not punishment

Acceptance Criteria: Child paints outside a region boundary → friendly message appears within 500ms; message disappears after 3s; no state loss

FR-10: AI Coloring Feedback & Scoring P1 — Should Have

AI-powered feedback system that scores completed artwork, provides encouragement, and suggests improvements.

- **Scoring engine:** Evaluate coloring based on coverage (% of regions filled), neatness (brush strokes within boundaries), creativity (color variety), and completeness
- **Score display:** Fun, gamified score (e.g., ⭐⭐⭐ out of 5 stars, or "Amazing Artist!" / "Color Explorer" / "Keep Going!")
- **AI suggestions:** Gentle, age-appropriate correction prompts (e.g., "The rocket's nose cone is still white — try coloring it!" or "Great use of colors! Try using more than 3 different colors next time")
- **Celebration animations:** Confetti / fireworks on high scores
- **Progress over time:** Track improvement across sessions; show "You're getting better!" trends in the parent dashboard
- **No negative feedback:** All messaging is positive and encouraging — "almost there!" instead of "wrong"
- **CBC alignment:** Scores map to CBC competency levels for teacher reports

Acceptance Criteria: Child completes a page → AI score appears with stars and personalized feedback; suggestions are age-appropriate; scores persist in profile

5.3 Phase 3 — Scale & Platform (Months 7–12)

Goal: Expand platform reach and revenue channels

FR-11: Mobile Apps (iOS & Android) P1 — Should Have

Native mobile apps wrapping the web experience with touch-optimized controls.

- React Native or Capacitor wrapper
- Touch gestures: tap-to-fill, pinch-to-zoom, swipe navigation
- Offline mode with local storage
- Push notifications for streaks and new content
- In-app purchases for subscriptions

Acceptance Criteria: App published on App Store and Google Play; feature parity with web

FR-12: Educator / Classroom Tools P1 — Should Have

Tools for teachers to use TechToons in classroom settings.

- Class creation with join codes
- Assign specific pages as "homework"
- View class-wide progress and individual reports
- Curriculum alignment tags (Kenya CBC — Digital Literacy strand)
- Bulk student account creation via CSV

Acceptance Criteria: Teacher creates class, assigns page, views class progress report

FR-13: Printable Worksheets P2 — Nice to Have

Generate print-ready PDF worksheets from any coloring page.

- "Print" button exports SVG as black-and-white PDF
- Includes page title and fun facts on the worksheet
- A4 and Letter size options
- Premium feature (Plus subscribers)

Acceptance Criteria: PDF downloads with clean line art and facts

FR-14: API & Partner Integrations P2 — Nice to Have

Open API for embedding TechToons in partner platforms.

- Embeddable widget (iframe) for partner sites
- API key authentication
- Custom branding options for partners
- Revenue share tracking

Acceptance Criteria: Partner embeds widget; usage tracked in admin panel

6. Non-Functional Requirements

REQUIREMENT	SPECIFICATION
Performance	First Contentful Paint < 1.5s on 3G; SVG coloring interaction < 50ms; optimized for low-bandwidth Kenyan connections
Accessibility	WCAG 2.1 AA compliance; keyboard navigation; screen reader support for UI controls
Security	Kenya Data Protection Act (DPA 2019) compliance; encrypted data at rest and in transit
Privacy	No third-party tracking for child accounts; GDPR/DPA data deletion on request
Scalability	Support 50K concurrent users; CDN-served static assets (Cloudflare with Nairobi PoP)
Browser Support	Chrome 90+ (dominant in Kenya), Samsung Internet, Firefox 100+, Edge 90+
Uptime	99.9% availability SLA
Internationalization	i18n framework in place; launch with English, then Swahili; future: French (for East Africa expansion)

7. Technical Architecture (Proposed)

💡 **Why Supabase?** Supabase provides an all-in-one backend (PostgreSQL database, authentication, file storage, edge functions) with a generous free tier (50K MAU, 500MB DB, 1GB storage). This drastically reduces backend complexity and cost — ideal for a Kenya-first MVP where every KES counts. For scale, Supabase Pro ($25/mo) handles the first 100K+ MAU before custom infrastructure is needed.

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          │ Client (Web/Android) │
          │ React 19 + TypeScript + Vite │
          │ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
          │ │ Coloring │ │ Auth │ │ Parent Dashboard │ │
          │ │ Engine │ │ (Phone#) │ │ + Analytics │ │
          │ └──────────┘ └──────────┘ └──────────────────┘ │
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          │ HTTPS / REST + Realtime
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          │ Supabase (Backend-as-a-Service) │
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          │ │ Auth │ │ Database │ │ Storage │ │
          │ │ (Email, │ │(PostgreSQL│ │ (Exported │ │
          │ │ Phone, │ │ Users, │ │ Artwork, SVGs) │ │
          │ │ Google) │ │ Progress)│ │ │ │
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          │ │ Edge │ │ Realtime │ │
          │ │ Functions│ │ (Sync) │ │
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          ┌──────────────────▼──────────────────────────────┐
          │ External Services │
          │ ┌──────────────────┐ ┌──────────────────────┐ │
          │ │ M-Pesa (Daraja) │ │ Cloudflare CDN │ │
          │ │ + Stripe Cards │ │ (Nairobi PoP) │ │
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8. Monetization Strategy

Revenue Streams

STREAM	MODEL	PROJECTED YEAR 1
TechToons Plus	KES 250/mo (~$1.99) per family	KES 3.6M (~$28,000)
TechToons Classroom	KES 500/mo (~$3.99) per school	KES 1.2M (~$9,500)
Printable Worksheets	Included in Plus	— (retention driver)
NGO/Govt Partnerships	Sponsored deployments (e.g., Safaricom Foundation, KICD)	KES 1.5M (~$11,500)
Branded Content Packs	Sponsored collections (e.g., Safaricom, M-Pesa, Konza Technopolis)	KES 1M (~$7,700)
	Projected Total	**KES 7.3M (~$56,700)**

Pricing Justification

- Priced for Kenyan families: KES 250/mo is below the cost of a single meal at KFC (KES ~600)
- Global competitors (Toca Boca at KES 650/mo, ABCmouse at KES 1,700/mo) are unaffordable for most Kenyan families
- **M-Pesa** is the primary payment method — frictionless for 90%+ of Kenyan adults
- Classroom pricing allows schools to fund from per-learner capitation grants (KES 1,420/pupil/yr)

Cost Structure

📋 **Note:** Detailed pricing and cost breakdown will be costed in collaboration with our software development partner. Key cost categories include: engineering, content creation, infrastructure, marketing, legal/compliance, and payment processing fees.

9. Competitive Analysis

FEATURE	TECHTOONS	TOCA BOCA	KYTABU	ENEZA EDUCATION	ELIMU
Tech/STEM Theme	✅ Unique	❌	❌	Partial	Partial
Coloring + Learning	✅	❌	❌	❌	❌
Web-Based (No Install)	✅	❌	✅	✅ (USSD+Web)	✅
CBC Alignment	✅	❌	✅	✅	✅
M-Pesa Payment	✅	❌	✅	✅	✅
Price (KES/mo)	KES 250	KES 650	KES 200	KES 10/day	KES 300
Target Age	4–10	3–9	6–18	6–18	3–12
Export Artwork	✅	❌	❌	❌	❌

Competitive Moat: TechToons is the only Kenyan product combining creative coloring with tech-specific STEM learning. Local competitors (Kytabu, Eneza, eLimu) focus on text-based subject tutoring — none offer a creative, interactive art experience. Global competitors (Toca Boca) are priced out of the Kenyan market and lack CBC alignment.

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10. Go-To-Market Strategy

Launch Plan (Kenya-First)

PHASE	TIMELINE	ACTIVITIES
Soft Launch (Nairobi)	Month 3	Launch web app with free tier; onboard 20 beta parents in Nairobi
School Pilots	Month 3–4	Partner with 5 Nairobi primary schools; align with CBC Digital Literacy strand
Social Media Push	Month 4–5	TikTok/Instagram Reels in Kenya; WhatsApp group outreach to parent communities
M-Pesa Integration	Month 5	Enable M-Pesa payments for Plus subscriptions
Education Outreach	Month 5–6	Present at KICD curriculum workshops; partner with Kenya Education Network (KENET)
Google Play Launch	Month 7	Android app (most Kenyan devices); Google Play Family category
County Expansion	Month 8–10	Expand to Mombasa, Kisumu, Nakuru, Eldoret
East Africa Scale	Month 11–12	Launch in Uganda, Tanzania, Rwanda

Distribution Channels

1. **Google Play Store** — Android dominates Kenya (95%+ mobile market share); Family category
2. **WhatsApp & Social Media** — WhatsApp groups are the primary parent community channel in Kenya; TikTok/Instagram for awareness
3. **School Partnerships** — Direct outreach to CBC-implementing schools; training for Digital Literacy teachers
4. **Safaricom/M-Pesa Ecosystem** — Explore M-Pesa Mini-Apps and Safaricom app store placement
5. **Education Organizations** — Kenya Institute of Curriculum Development (KICD), KENET, UNESCO Nairobi
6. **Organic Search** — SEO for "CBC digital literacy games Kenya", "kids STEM activities Nairobi"

11. Risks & Mitigations

RISK	LIKELIHOOD	IMPACT	MITIGATION
Kenya DPA 2019 compliance	Medium	Critical	Engage Kenyan legal counsel; data stored in Africa-region servers
Low conversion / willingness to pay	High	High	Strong free tier; KES 250 pricing validated against ARPU benchmarks; M-Pesa removes friction
Internet connectivity gaps	Medium	Medium	Lightweight SVGs (~7KB each); service worker caching for offline use
Content creation bottleneck	Medium	Medium	SVG illustration pipeline; hire Kenyan freelance illustrators via platforms like Brighter Monday
Device fragmentation	Medium	Medium	Test on low-end Android devices common in Kenya (Tecno, Infinix, Samsung A-series)
School IT infrastructure	Medium	Medium	Web-based (no install needed); works on Chromebooks and shared school tablets
Competitor copies STEM angle	Low	Medium	First-mover advantage in Kenya's creative EdTech space

12. Roadmap Summary

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Month 1–3 ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓ Phase 1: Foundation
          ├─ User accounts & auth
          ├─ Cloud save
          ├─ Expand to 20+ pages
          └─ Freemium + Stripe

          Month 4–6 ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓ Phase 2: Engagement
          ├─ Learning sidebars
          ├─ Achievements & badges
          ├─ Parent dashboard
          ├─ Brush boundary guidance
          ├─ AI scoring & feedback
          └─ Community gallery
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Month 7-12 ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Phase 3: Scale
├─ iOS & Android apps
├─ Classroom tools
├─ Printable worksheets
└─ Partner API
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13. Appendix

A. Glossary

TERM	DEFINITION
Region	An individual SVG path element that can be filled with color
Collection	A themed group of 5 coloring pages
Streak	Consecutive days a child uses the app
CBC	Competency-Based Curriculum — Kenya's national education framework (2017+)
DPA 2019	Kenya Data Protection Act — governs collection of personal data including children's
M-Pesa	Mobile money platform by Safaricom, used by 90%+ of Kenyan adults
KICD	Kenya Institute of Curriculum Development — the body overseeing CBC implementation
KES	Kenyan Shilling (1 USD ≈ 129 KES as of Feb 2026)

B. Open Questions

1. Should we add Swahili translations from launch or wait for user demand data?
2. Can we integrate with the government's Digital Literacy Programme (DLP) devices in schools?
3. Should we partner with Safaricom Foundation for a sponsored free tier in rural schools?
4. What's the optimal M-Pesa payment interval — monthly or weekly (e.g., KES 65/week for lower commitment)?
5. Should we include Kenya-specific illustrations (e.g., matatu, Kenya satellite, M-Pesa phone) in future collections?

This document is a living artifact. It will be updated as business requirements evolve and user feedback is gathered.

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